

The mathematical bridge connecting the observed, unlinked marginals (Panel A) and the recovered latent SGPC regime (Panel C) has two ingredients:

1. A baseline copula C_0 providing a dependence template from prior reference-quantiles U to current reference-quantiles V .²
2. A stochastic growth regime H that governs how subgroup S allocates conditional percentiles within that template.^{3, 4}

Together, C_0 and H define a *forward map* from a candidate regime to a *predicted* current-grade marginal CDF. Panel B2 visualizes this map via the induced $F_H(v)$, and Panel B1 *inverts* it by choosing the regime whose F_H best matches the observed current marginal $F_{V,S}^{\text{obs}}$.

Let C_0 be a baseline copula. C_0 induces the conditional CDF (transition kernel)

$$F_0(v | u) = \Pr(V \leq v | U = u) = \frac{\partial}{\partial u} C_0(u, v).$$

Its inverse in v (the conditional quantile kernel) is

$$Q_0(p | u) = F_0^{-1}(p | u).$$

Intuitively, for each starting quantile u , the kernel $F_0(\cdot | u)$ describes the distribution of possible ending quantiles V under the baseline dependence structure C_0 .

Let $P \in [0, 1]$ be a random variable (a latent SGPC on the 0–1 scale) and let p denote a realization in $[0, 1]$. A *growth regime* H is the distribution of P , represented by its CDF

$$H(p) := \Pr(P \leq p), \quad p \in [0, 1].$$

Given $U = u$ and a draw $P = p$, the model generates an outcome percentile by:⁵

$$V = Q_0(P | U = u).$$

The uniform regime $H(p) = p$ (i.e., $P \sim \text{Uniform}(0, 1)$) corresponds to occupying the baseline kernel with no tilt. Departures from $H(p) = p$ tilt the growth regime mass toward lower or higher conditional percentiles relative to the baseline regime.

Although we do not observe linked pairs (u, v) in TIMSS- or NAEP-style settings, the model implies a forward map from the observed prior marginal F_U to the predicted current marginal F_H . For any fixed u ,

$$\begin{aligned} \Pr(V \leq v | U = u) &= \Pr(Q_0(P | u) \leq v) \\ &= \Pr(P \leq F_0(v | u)) \\ &= H(F_0(v | u)). \end{aligned}$$

Therefore, by the law of total probability, the subgroup's predicted current-grade marginal CDF is

$$\begin{aligned} F_H(v) &= \Pr(V \leq v) \\ &= \mathbb{E}_U [H(F_0(v | U))] \\ &= \int_0^1 H(F_0(v | u)) dF_U(u) \\ &\approx \frac{\sum_{i=1}^n w_i H(F_0(v | u_i))}{\sum_{i=1}^n w_i}, \end{aligned}$$

where $\{u_i, w_i\}$ are the observed prior pseudo-observations and survey weights for the subgroup (i.e., a weighted empirical approximation to F_U).

We estimate the subgroup's growth regime by searching over an admissible class $\mathcal{H}$ of candidate CDFs on $[0, 1]$. Panel B1 shows the heat map of $\mathcal{H}$ over the *Beta regime family*

$$\mathcal{H}_{\text{Beta}} = \{H_{m,\kappa}(\cdot) : P \sim \text{Beta}(\kappa m, \kappa(1 - m)), 0 < m < 1, \kappa > 0\},$$

where $m = \mathbb{E}[P]$ controls the regime center (mean SGPC on the 0–1 scale) and κ controls concentration (dispersion of latent SGPC).⁶

For each candidate $H \in \mathcal{H}$, Panel B2 provides the induced *predicted* current-grade marginal

$$F_H(v) = \mathbb{E}_U [H(F_0(v | U))].$$

The growth regime estimate for subgroup S is defined by the minimum-distance problem

$$\hat{H}_S = \underset{H \in \mathcal{H}_{\text{Beta}}}{\text{argmin}} W_1(F_{V,S}^{\text{obs}}, F_H),$$

where

$$W_1(F, G) = \int_0^1 |F(v) - G(v)| dv$$

is the Wasserstein-1 (earth mover's) distance between the observed and predicted current-grade marginal CDFs.⁷

The heat map displays $\log_{10}$ Wasserstein-1 distance over $(m, \log_{10} \kappa)$: lighter regions indicate candidate regimes whose induced F_H more closely matches $F_{V,S}^{\text{obs}}$. The marked optimum corresponds to $\hat{H}_S$ (equivalently, $(\hat{m}, \hat{\kappa})$). For reference, the uniform regime $H(p) = p$ occurs at $m = 0.5$ and $\kappa = 2$ (i.e., $\text{Beta}(1, 1)$), indicated by the $U(0, 1)$ marker.

The baseline copula, C_0 , combined with the best fit growth regime estimate, $\hat{H}_S$, induces the Wasserstein-1 minimized CDF, $F_{\hat{H}_S}(v)$, and its density, $f_{\hat{H}_S}(v)$, which are shown in panel B2 and A, respectively, with a dashed-black line.